

Advanced VLSI Design - Lab 2

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The purpose of this lab is to combine the 5-bit DAC completed in Lab 1 with an 8-pin pad-frame. The following sections cover the layout and schematic designs of the pad-frame as well as briefly covering the 5-bit DAC. The final section demonstrates the integration of the DAC with the pad-frame.

1 Design and Layout of a Pad-frame

1.1 Layout and Schematic of a Single Pad

The schematic of the pad simply used an Off-page node with the export **inout**. A corresponding icon was made using the icon view tool and can be seen in figure 1.

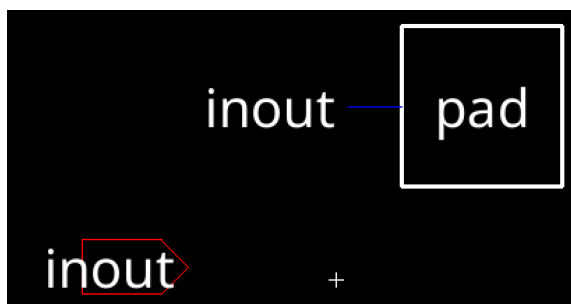


Figure 1: Single Pad Schematic with Corresponding Icon

The pad layout began with a 400x400 square found in the artwork category of components. This square outline is designed to represent spacing between the different pad-frames. A 250x250 metal-2 - metal-3 contact was centered within the drawn square, and a 200x200 overglass layer was placed using the **Passivation-node** in the **Pure** functions within the *mocmos* tab of components. Similarly, the export **inout** was used. The resulting single pad can be seen in figure 2.

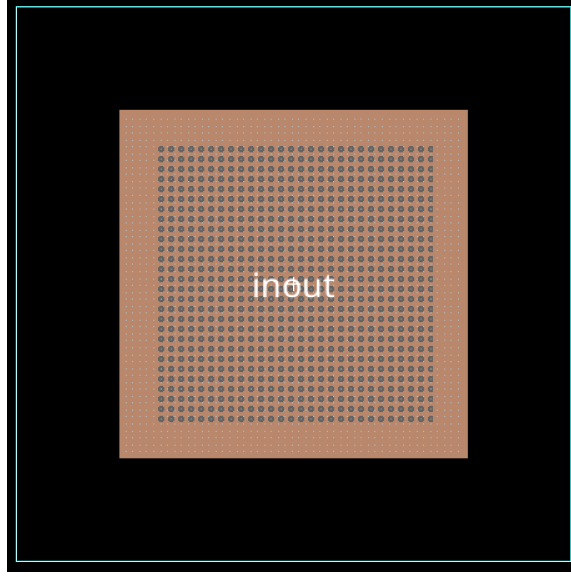


Figure 2: Single Pad Layout

1.2 Layout and Schematic of an 8-Pin Pad-Frame

The schematic for the 8-pin pad-frame was made with a similar strategy used to create the pad. An Off-page node was first placed with the export name **pin[1:8]** followed by the icon created for the singular pad-frame. The node and icon were connected via an arc bus to symbolize the connection of multiple electrical signals. A resulting icon was made from the new schematic using the make icon view tool. Both the icon and schematic can be seen in figure 3.

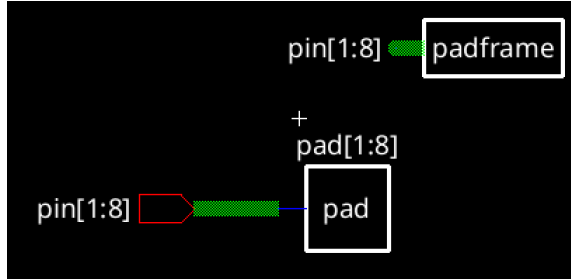


Figure 3: Pad-Frame Schematic with Corresponding Icon

To create the 8-pin pad-frame, the singular pad layout was dragged-and-dropped into the new pad-frame layout. The pad-frame was created by using the edit $\rightarrow$ array tool in Electric, with the x-repeat factor and y-repeat factor set to 4x4. The result of this is a full 4x4 square of pads; thus, to get the desired pad-frame, the outer corners were deleted along with the inner four pads. As a note, creating an array duplicates layouts on top of each other, so to ensure the correct pads

were receiving the correct export names, "top-layers" were deleted from the layout. The resulting pad-frame can be seen in figure 4, with a corresponding 3D render seen in figure 5.

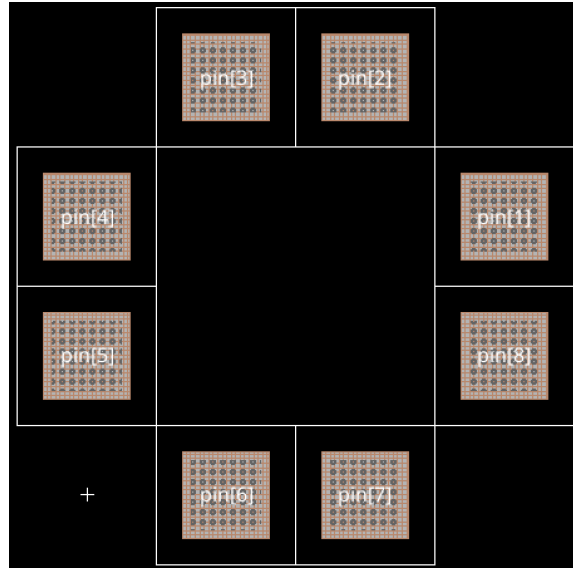


Figure 4: 8-Pin Pad-Frame Layout

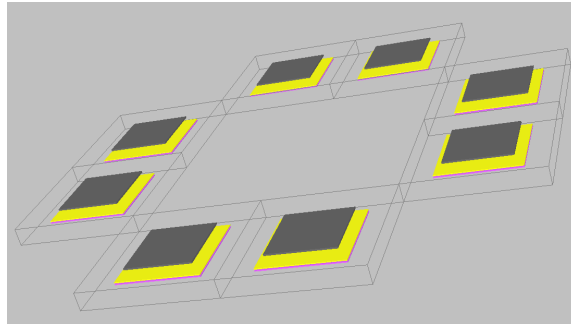


Figure 5: 3D Render Corresponding to the 8-Pin Pad-Frame Layout

2 5-Bit DAC

The following sub-sections cover the making of a 5-bit DAC that was placed into the pad-frame and were taken directly from VLSI Lab 1 - 5-Bit DAC.

2.1 Schematic and Layout of a Simple Voltage Divider

A simple 2R-R voltage divider was made using three $10k\Omega$ resistors. Utilizing a set width of 15 and $R_{square} = 855 \frac{\Omega}{square}$, the length could be calculated to approximately 175:

$$R = R_{square} \cdot \frac{L}{W} \Rightarrow L = \frac{R \cdot W}{R_{square}} \Rightarrow L = \frac{10k\Omega \cdot 15}{855 \frac{\Omega}{square}} \approx 175\Omega$$

Three Off-page nodes were used, with export names labeled **in1**, **out1**, and **bot1**. A corresponding icon was made using the make icon view tool. The schematic and icon for the 2R-R voltage divider can be seen in figure 6.

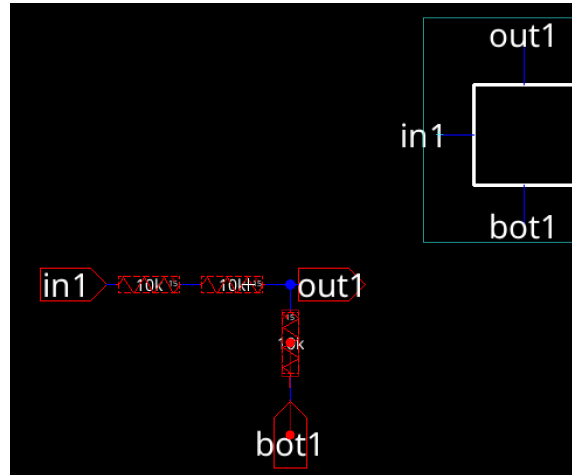


Figure 6: 2R-R Voltage Divider and Corresponding Icon

To create the layout for the 2R-R voltage divider, three n-well resistors were used, and the lengths and widths were set to their corresponding values. The wired 2R-R voltage divider layout can be seen in figure 7.

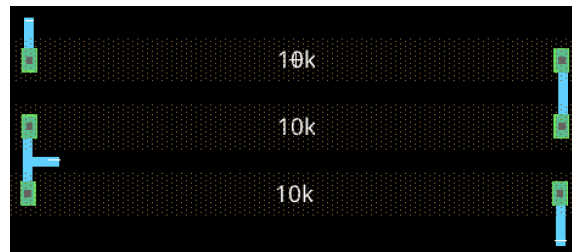


Figure 7: 2R-R Voltage Divider Layout

2.2 Layout and Schematic of a 5-Bit DAC

To build the schematic for the 5-bit DAC, the 2R-R voltage divider icon was brought into the new schematic window and copied over for a total of 5 icons. Off-page nodes were applied to each voltage divider and export labeled according to their corresponding bit inputs. The schematic for the 5-bit dac with corresponding icon and LTspice simulation code an be seen in figure 8.

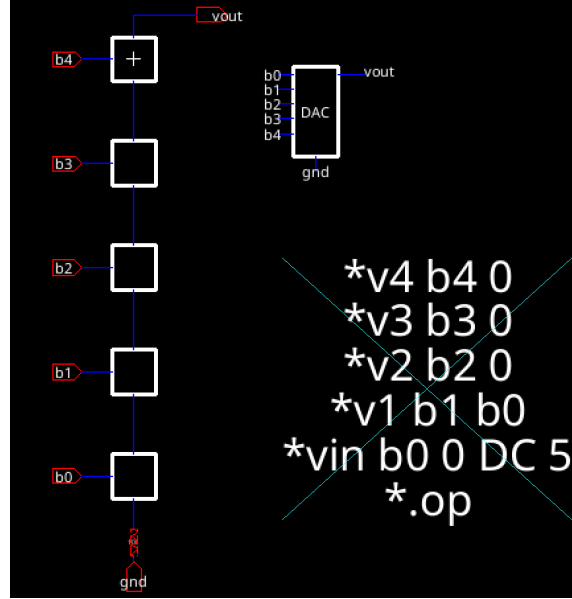


Figure 8: 5-bit DAC Schematic with Corresponding Icon and LTspice Code

To build the layout for the 5-bit DAC, an individual 2R-R voltage divider layout was placed to the new layout window and copied over for a total of five voltage dividers. The voltage dividers were wired together and each input and output was labeled accordingly. The layout for the 5-bit DAC and corresponding 3D render can be seen in figure 8 and 9.

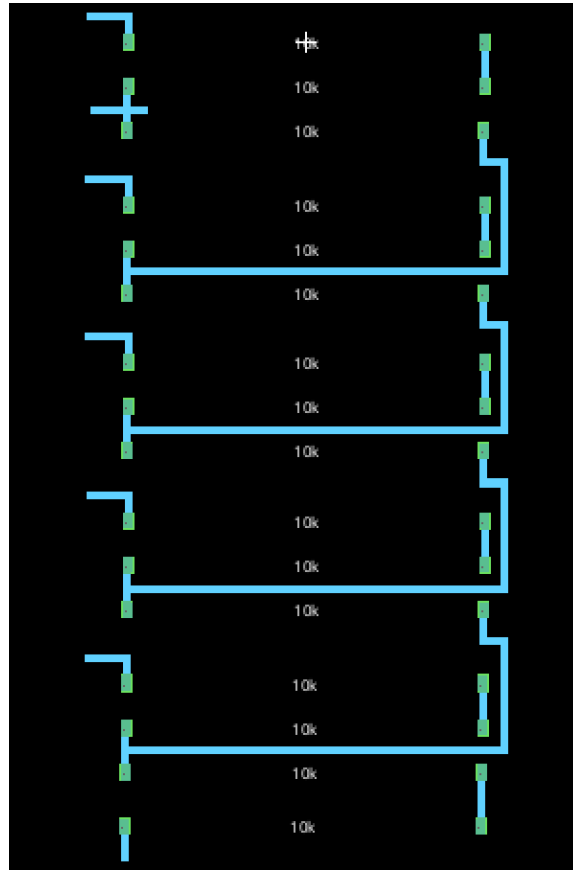


Figure 9: 5-Bit DAC Layout

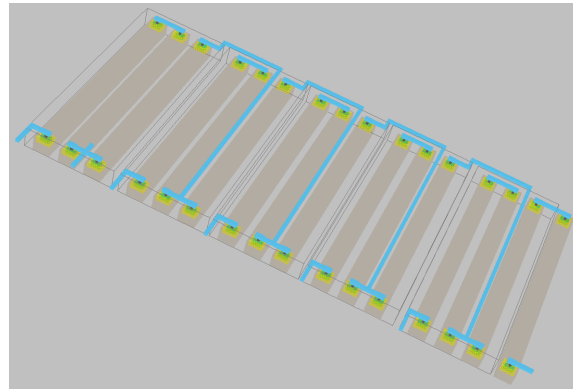


Figure 10: 3D Render of 5-Bit DAC

3 Connecting the DAC to the Pad-Frame

The final IC design started with creating a schematic using the 5-bit DAC icon and labeling pins according to their designated pads. The order for the pin to pad connection goes as follows:

Pin	Pad
b0	pin[1]
b1	pin[8]
b2	pin[7]
b3	pin[6]
b4	pin[4]
ground	pin[2]
v_{out}	pin[5]

Table 1: Caption

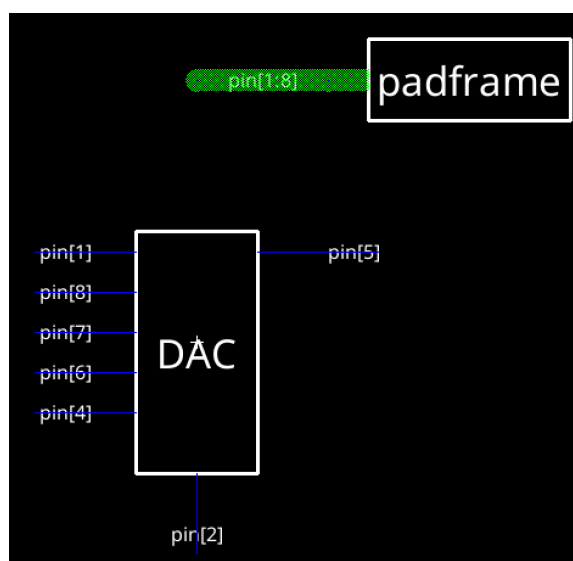


Figure 11: Final IC Schematic

To create the layout, the 5-bit DAC and pad-frame layouts were placed into the new layout view. As an important note, the pad-frame had to be set to 'hard to select' in order to be able to select pins and make proper connections. To allow for connections between the 5-bit DAC and the pad-frame, metal-1 to metal-2 connections were placed in front of each pad being used. Since the software does not automatically recognize connections being made from different metal layers, the *Toggle Special Select* tool was used to make traces between the pins and pads. The final IC layout can be seen in figure 10 with its corresponding 3D render in figure 11.

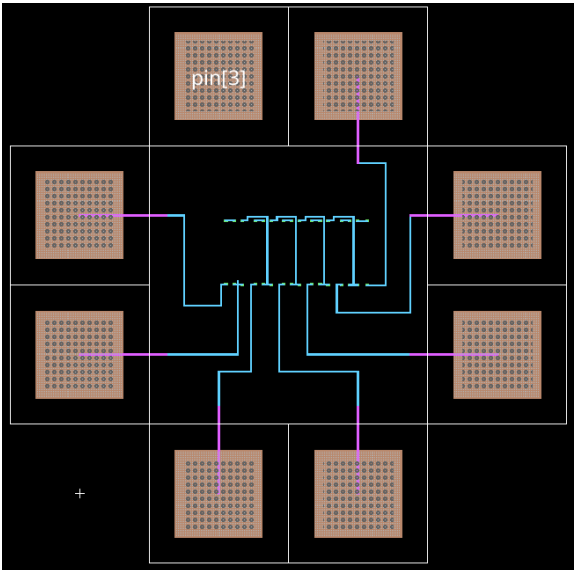


Figure 12: Final IC Layout

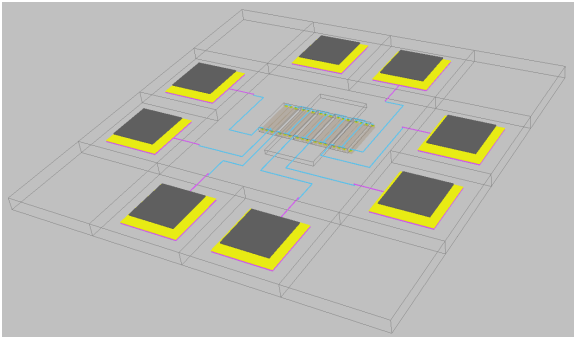


Figure 13: 3D Render of Final IC Design